Auxiliary-input leads 2026-11-12

Auxiliary inputs, curves and points: lead chase and five follow-ups (2026-11-12)

Question. What is the current state of every lead about auxiliary curves, auxiliary points and Cheon's DLP with auxiliary inputs, and how far can the five follow-ups the owner approved on 2026-11-12 be taken from this session? **Class.** Desk work and literature only. Nothing was run, launched or measured. Every statement is `[pcc:unverified]`. **Report PDF:** report.pdf, built from this file. **Figures:** figures/lead_map.svg (lead state) and the recorded exponent graph of the wide study, ../cheon_ecbench_wide_20261106/alpha.svg.

Sources checked

- research-kb: **unavailable** (no MCP server connected; `scripts/research-kb` finds no catalog in this container).
- Lead atlas: `crypto-autoresearcher` cloned at `8c82ddc8` (2026-11-11). Every atlas file is AES-SIV vault ciphertext (`VLT1`); `KEY` is not set, so `tools/lead_atlas.py` cannot run. Lead fields were read from the in-repo copies: `research/lead_sweep_20261110/leads.json` and per-lead `RESULT.md`, `research/prime_field_lead_assessment_20261107/atlas_survey.md`, `papers/ecdlp-index-calculus-treatise/survey/generic.md` §19.
- Studies: `research/cheon_ecbench_wide_20261106/`, `cheon_ecbench_medium_20261106/`, `cheon_p224_shape_20261022/`, `torsion_auxiliary_inputs_20260918/`, `srs_auxiliary_census_20260927/`, `pkm_supersingular_aux_20261015/`, `pkm_isogeny_tower_p192_20261015/`, `pkm_matched_prime_20261106/`, `pkm_falldegree_unblock_20261107/`; `research/notes/ecdlp-general/RESEARCH_TORSION_AUXILIARY_INPUTS.md`.
- Open PRs: #3932 (pooled Cheon intervals), #3938 (13 new leads).
- Literature: Kiltz–Winterhof, *Bull. Austral. Math. Soc.* 69 (2004) 305–315 (read in full); Lange–Winterhof COCOON'02 and AAECC'03 (metadata only, paywalled). See LITERATURE_LANGE_WINTERHOF.md.

1. Lead state

Auxiliary inputs, curves and points: lead state on 2026-11-12

every status [pcc:unverified]; atlas read from the in-repo copies (vault unreadable)

Given leaked powers [α^i]G (Cheon)

LEAD-RHO-cheon-aux-acquisition — closed (theorem)
powers not obtainable from G,[α]G: plain ECDLP untouched

Cheon comb vs matched rho — measured, separated
cheon_ecbench_wide_20261106: α 0.455 vs 0.508 (n−1), 0.432 vs 0.510 (n+1), 36–56 bits; PR #3932 intervals

LEAD-RHO-cheon-toy-reproduction — open
EXP-AUXIN-25a29a approved, not run: spec in the vault

LEAD-RHO-cheon-deployed-applicability — blocked
revised protocol (H-AUXIN-6db354) in the vault, unread

LEAD-EXO-hnp-advice-leakage — open
toy: slope 0.25 vs BSGS 0.50 given the leak

LEAD-RHO-cheon-value-set-embedding — stub
Cheon–Kim τ_f: small value set for d ∤ r±1, deg < r^{1/3}?
open in the literature; not filed (novelty check partial)

Auxiliary curve / point as the object

Elliptic / torsion embedding (Kim–Cheon) — closed
label is a quotient φ_d/ψ_d², not formable in the exponent: honest cost ≈ r/d (500–12,000× rho at 14–20 bits)

LEAD-EXO-lange-winterhof-interpolation — open
neighbour theorems recorded; COCOON'02 text paywalled

LEAD-TRF-same-field-genus-g-embedding — open
toy confirms Θ(p) cofactor obstruction

LEAD-ICP-pkm-tower-algebraic-oracle — blocked
stage C: solving degree grows in m (6/6 ladders);
t = 13 draw 0 lost 3× to Spot; non-Spot not allowed

Supersingular vs ordinary aux curve — settled
same D; aux work 1–2 % of S; ≈ plain base

closed / settled · measured result · open, next step · blocked · proposed (stub, not in atlas)

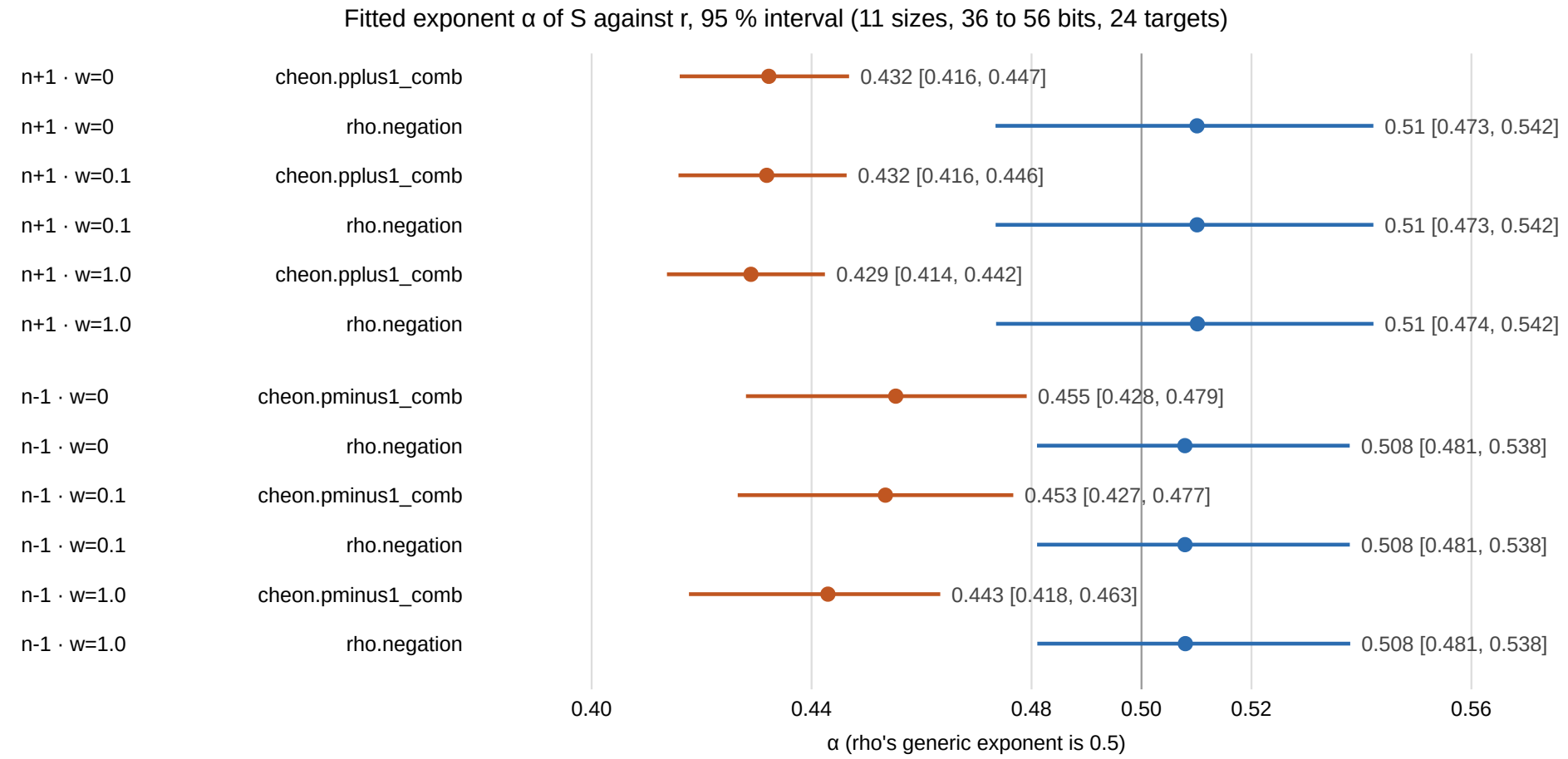
Lead map

lead	disposition	latest evidence
`LEAD-RHO-cheon-aux-acquisition`	closed_obstructed (theorem)	`[α²]G` from `G, [α]G` is square-DH. The 2026-11-10 sweep toy was inconclusive (positive control and A/A failed).
Cheon comb vs matched rho, powers given	measured	Wide study: α 0.455 [0.428, 0.479] vs rho 0.508 [0.481, 0.538] (n−1), 0.432 [0.416, 0.447] vs 0.510 [0.473, 0.542] (n+1), w = 0, 36–56 bits, 1,056 replayed runs. Pooled ratio comb/rho 0.525 [0.473, 0.584] (n−1), 0.367 [0.318, 0.422] (n+1) in open PR #3932.

lead	disposition	latest evidence
`LEAD-RHO-cheon-toy-reproduction`	open_next_step	Sweep toy matched `1.5(√(r/d)+√d)` within 4 %; inconclusive because one negative-control cell sat at exactly 5 %. EXP-AUXIN-25a29a not run (§2, item 1).
`LEAD-RHO-cheon-deployed-applicability`	open_blocked	Revised protocol for H-AUXIN-6db354 awaits approval (§2, item 4). Powers-of-tau census: 5.74–14.32 bits.
`LEAD-EXO-hnp-advice-leakage`	open_next_step	Sweep toy: Cheon slope 0.2516 vs BSGS 0.5039 given the leak; its rho baseline is mis-specified (slope 1.006), so its rho ratios are void.
Elliptic / torsion embedding (Kim–Cheon)	closed by measurement; no atlas lead	Label is a quotient; honest cost ≈ `r/d`.
`LEAD-EXO-lange-winterhof-interpolation`	open_next_step	Neighbour theorems recorded (§2, item 3).
`LEAD-TRF-same-field-genus-g-embedding`	open_next_step	Sweep toy confirms `#Jac = #E₁·#E₂` on 132 curves, the Θ(p) cofactor obstruction.
`LEAD-ICP-pkm-tower-algebraic-oracle`	open_blocked	Stage C: solving degree grows in m on 6 of 6 ladders. Matched primes: D follows t, not p, and equals the null. t = 13 undecided (§2, item 5).
Supersingular vs ordinary auxiliary curve	settled	Same D in 14 cells; auxiliary work 0.9–1.9 % of S vs 10–27 % searched; against a plain base unresolved or 1–7 % dearer.
Cheon–Kim value-set embeddings	**no lead**; stub written	§2, item 2.

A conflict resolved. The 2026-09-18 note's Cheon exponent 0.23 and the ecbench exponents of about 0.45 are both consistent with theory: the first used `d ≈ √r` (predicted 1/4), the ecbench studies use P-224-shaped `d ≈ r^0.129` and `r^0.211` (predicted `(1 − ρ)/2` ≈ 0.435 and 0.394). Different `d`, not a contradiction.

A retraction to carry. The Cheon/rho ratios of `research/cheon_p224_shape_20261022/` were withdrawn on 2026-10-23 (H3 gate failed); the wide study supersedes them.



Fitted exponents, wide study (recorded values, 95 % intervals)

2. The five follow-ups

item	outcome	why / what was done	what unblocks it
1. Run EXP-AUXIN-25a29a (H-AUXIN-a86953: 56-bit r, d = 65536, stage counts, gain fractions)	**not run**	Its frozen card lives only in the vault. Only a 100-character excerpt is readable. Running a reconstruction under the approved experiment's id would not be that experiment.	The vault KEY (or the card in plaintext), then the run on Iso Lab EKS with a resume checkpoint and a `bench.record/v1` record.
2. File a lead for value-set / auxiliary embeddings	**stub written, not filed**	LEAD_STUB_value_set_embedding.md. Novelty check covered the 173-lead sweep index, PR #3938 and full-text search of `crypto` and `cryptanalysis`.	A vault and research-kb novelty check, then filing in the atlas.
3. Lange–Winterhof exact theorem	**partly done**	Kiltz–Winterhof 2004 Theorems 5 and 6, stated as extending Lange–Winterhof Theorem 3, are transcribed with scope in	Library access or an author copy.

item	outcome	why / what was done	what unblocks it
		LITERATURE_LANGE_WINTERHOF.md. The COCOON'02 text is still paywalled.	
4. Approve the revised deployed-applicability protocol	**owner approval recorded in principle, not bound**	The owner said yes on 2026-11-12. The protocol text for H-AUXIN-6db354 is in the vault and was not read, so the approval cannot be bound to a document hash from here.	Whoever can open the vault records the approval against the protocol's hash.
5. Re-launch PKM `pfd-x-t13-k0-c6` on non-Spot capacity	**blocked by policy**	`taskq/deploy/eks-karpenter/spot-only-policy.yaml` admits only Spot NodePools; `COST_CONTROLS.md` forbids an On-Demand fallback; the CRIU resumable route is qualified to an 8 GiB envelope (the cell needs 56 GiB); this container has 15 GB. Three Spot attempts were already lost (`SPOT_LOSSES.md`).	An administrative exception for one On-Demand node (policy change), or an owner decision for a fourth Spot attempt, or a memory-qualified CRIU route.

3. Next steps

1. Supply the vault key to a session (or export the two AUXIN records); then items 1, 2 and 4 can proceed as written.
2. Decide item 5: a one-off On-Demand exception, a fourth Spot attempt, or accept "undecided at t = 13".
3. Read Cheon–Kim 2016 §§3–5 for the value-set stub's next step (desk).